

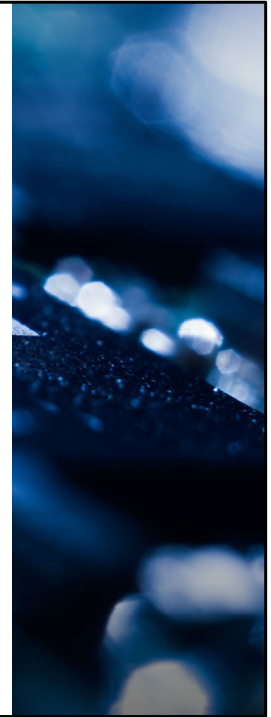


SC1006

Computer Organisation and Architecture

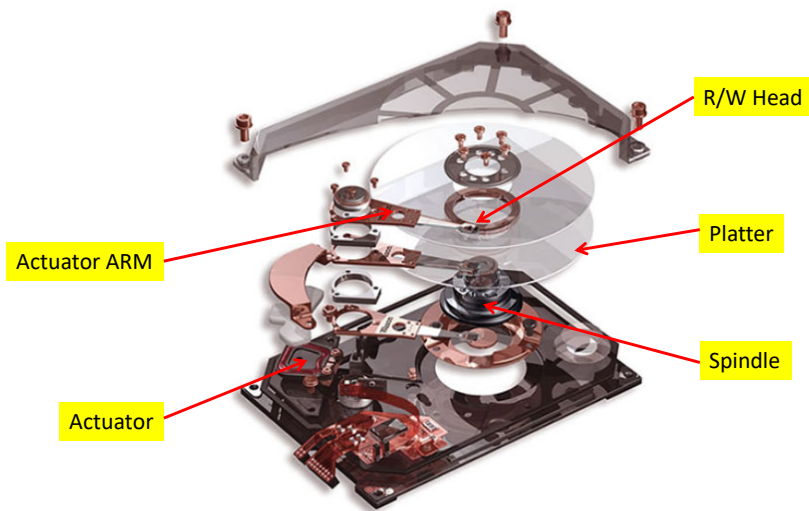
HDD & SSD

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- This section is an introduction on HDD & SSD

Magnetic Hard Disk



Source: <https://www.backblaze.com/blog/hdd-versus-ssd-whats-the-diff/>

This diagram shows a hard disk.

The Actuator ARM which carries the R/W head around different parts of the HDD media

The Actuator is an assembly of strong magnets and electrical windings. The resultant force produced when electric current flow through the wires is used to propel the actuator ARM.

The R/W head which picks up the changes in magnetic field on the media and converts these fluctuation into data bits.

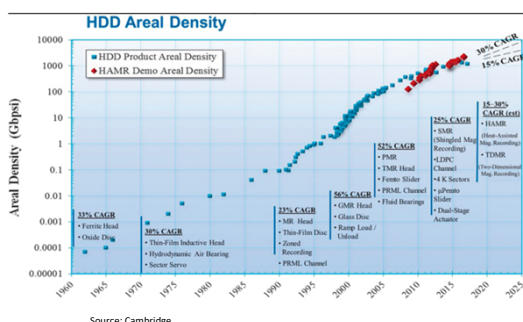
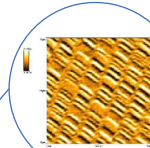
The Platter which is the storage media. Each Platter consists of two surfaces, top and bottom surfaces. These day, data is stored on both surfaces.

Lastly, the spindle which is a motor used to spin the platter at very high speed of up to 15K rpm.

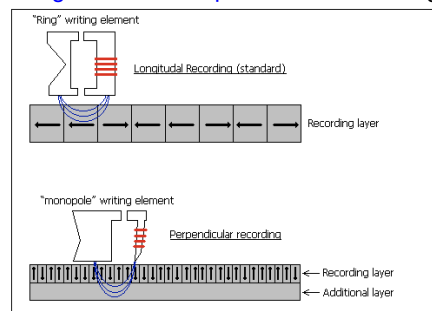
Magnetic Hard Disk



[Source] <https://www.digitaltrends.com/computing/blackblaze-hard-drive-report-seagate-western-digital/>



Longitudinal vs Perpendicular Recording



- Stores data by magnetizing a thin film of **ferromagnetic media** on the circular disk known as **platter**.
- Video on Introduction to Hard Disk.
<https://www.youtube.com/watch?v=kdmLv1n82U>

Magnetic hdd is still the main mass storage for Data Centers, Servers, and Desktop PC. However, it is fast being replaced by semiconductor memories such as SSD in the notebook products.

Magnetic HDD stores data by changing the orientation of the tiny magnetic elements in the circular disc platter.

There are Two types of recording method: Longitudinal and the more advanced perpendicular recording.

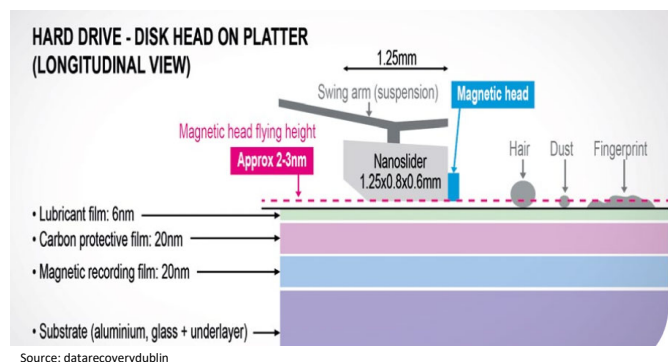
Difference between these two types of recording is in the orientation of the magnetic elements: as the name implies, longitudinal is along the surface of the platter while the other is perpendicular to the surface (which offers higher data storage density) The different orientation of the magnetic element will be interpreted as ones and zeroes by the RW head.

On the bottom left corner, we have a plot of the HDD areal density vs time. The graph shows how hard the HDD maker is working to enable a continuous improvement in HDD technology. Which is essential in enabling HDD to continue to be the memory element used in huge storage use cases such as the Data Centers.

HDD Technology



- The magnetic head uses the **wind pressure** generated by the high speed rotation of the disk to fly above the disk surface, similar to the principles employed by the aircraft.
- Fly height is lower than a finger print mark.



Flying the Magnetic head is often compared to flying a jumbo jet.

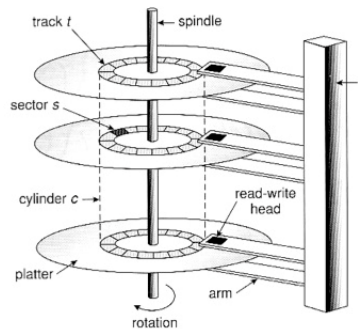
On the web, you'll find many different figures when people made this comparison. This is because it was done at different timeline, the technology they reference to is therefore different. And since HDD technology changed so quickly, you'll find that the number quoted differ by a large amount.

There is one source quoting that flying the RW head is like flying the jumbo jet a few mm above the ground.

With the actual height the RW head flies, even a finger print mark is taller than the fly height and will crash the RW head.

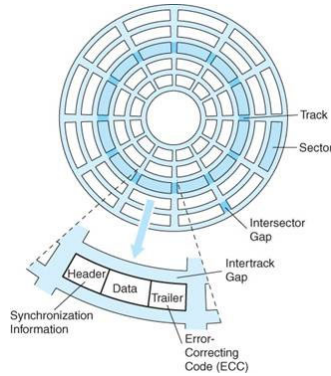
Bottomline, it shows the huge amount of technology and precision that has gone into the tiny HDD.

Magnetic HDD Data Organization



[Source] https://www.researchgate.net/figure/Sector-and-Platter-Design-1_fig11_303945698

- Computers often use magnetic hard disks for large secondary storage devices.
 - One or more **platters** on a common **spindle**.
 - Platters are covered with thin magnetic film.
 - Platters rotate on **spindle** at constant rate.
- **Data** is stored on the **surface** of the platter in concentric rings called **tracks**.
 - Gaps between tracks to minimize interferences from adjacent tracks.
- **Tracks** are divided into **sectors**.
 - Minimum data block size is a sector.
- Stored data consists **header, data and trailer**.



[Source] http://www.edwardbosworth.com/My5155_Slides/Chapter12/DiskBasics.htm

Now let's go thru some definition of parts in detail.

HDD stores its data in the circular disk known as platters.

The reading and writing of info go thru the read/write head. There is one head to each surface of the platter.

RW head is attached to an actuator arm whose motion is initiated by electromagnetic forces.

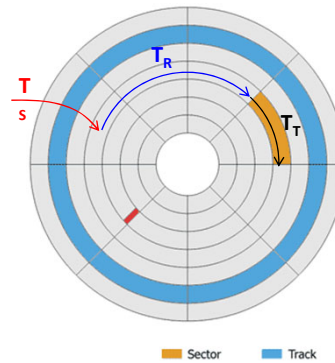
In terms of data organisation, the smallest container is known as a sector.

Multiple sectors will form a track, which is basically a circular track around the disc.

Within each sector, the data is formatted to provide synchronisation info, actual data payload and error correction code.

HDD Transfer Rate

- **Seek time (T_S)**
 - Time taken for the head to move to the correct track.
- **Rotational Delay (T_R)**
 - Time taken for the disk to rotate until the read/write head reaches the starting position of the target sector.
- **Access Time (T_A)**
 - Time from request to the time the head is in position ($T_S + T_R$)
- **Transfer Time (T_T)**
 - Time required to transfer the required data after the head is positioned.



Now for some definition of timing parameters used in HDD performance measurement.

First the Seek Time

This is the time it takes for the ARM to move from its resting position to the track that contains the target sector to be read.

The HDD vendor will typically supply a parameter known as the Track-to-Track seek time, since each movement from one track to another really takes different time, the parameter supplied by the HDD vendor is the average seek time to move from one track to another. That is, it is a statistical average value based on some test cases.

Then we have the Rotational Delay

Which is the time it takes to move to the start of the targeted sector, after the RW head reaches the correct track.

Adding the Seek Time and Rotational Delay will give you the Access Time.

Lastly, we have the Transfer Time, which is the time it takes to read the target data

after the RW head reached the start of the sector.

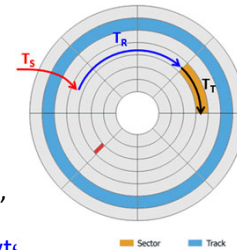
We discuss these 4 parameters in the following slides.

HDD Transfer Rate

- T_R (Rotational Delay) is dependent on the rotational speed, Revolutions Per Minute (RPM), of the disk. For calculations, RPM is usually converted to Revolutions Per Second (RPS). i.e. $RPS = RPM/60$
- For a random section, average rotational delay $T_{R,AV}$ may be calculated as

$$T_{R,AV} = \frac{0.5}{RPS} \text{ seconds}$$

- T_T (Transfer Time) is dependent on the rotational speed of the disk, the Track Density D_T (number of sectors per track), Sector Density D_S (number of bytes per sector) and the number of bytes N for the transfer.



$$T_T = \frac{N}{RPS * D_T * D_S}$$

Rotational delay depends on the rotation speed. The faster the rotation speed, the shorter the rotational delay.

Assuming we start from any random section, then the following formula will give the average rotational delay.

So how did we derive this formula?

Since the RW head could land on any sector when it reaches the correct track, Under the best case scenario, the RW head is just above the target sector, which means that we have zero delay.

Under the worst case scenario, the RW head had just past the target sector, which means that it need to wait for a full revolution of the disk before it can read the data (since the RW head rotates in one direction only). In this case the delay is equal to time it take to rotate one round, which is $1/RPS$.

So if we take the average, it'll be $0.5/RPS$, the formula given in the slide.

For the formula on Data Transfer Time,

The assumption here is that the RW head will be able to read the data once it passes over the corresponding sectors, so transfer time is equal to the time

needed to transverse the targeted sectors.

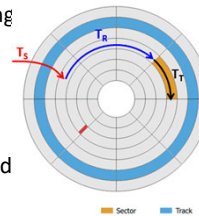
$Dt \cdot Ds$ gives the total number of bytes per track and N is the number of bytes to read.

So $N/(Dt \cdot Ds)$ gives you the ratio of data needed vs total data in one track.

Since the angular velocity of the disk is a constant, this ratio multiplies with the time it take to revolve one round, which is $1/RPS$, gives the expression shown here for Tt .

HDD Transfer Rate Example

- A magnetic hard disk rotates at **15000 RPM**, with the following properties:
 - Average Seek Time, $T_s = 4\text{ms}$
 - Track density, $D_T = 500$ sectors per track
 - Sector density, $D_s = 512$ bytes per sector
- Calculate the **total time T_{TOTAL}** it takes to read a **3 KB file** stored **consecutive sectors** on the same track?



[Solution]

$\text{RPS} = 15000/60 = 250$ per second

Access time $T_A = T_s + T_r = 4\text{ ms} + (0.5/250) = 6\text{ ms}$

Transfer time $T_T = 3072/(250*500*512) = 48\text{ }\mu\text{s}$

Total time, $T_{\text{TOTAL}} = T_A + T_T = 6.048\text{ ms}$

Notice $T_A \gg T_T$. This example shows that accessing file whose data is distributed across sectors on different tracks on the magnetic hard disk would potentially incur more time.

Let's look at one work example to apply what we have learnt so far. With a HDD having the parameters shown in the slide, what is the total time taken of read a 3KByte file stored in consecutive sectors?

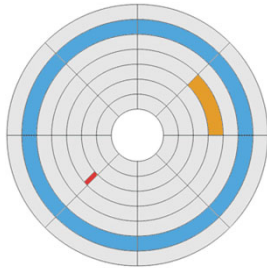
From the previous slide, we know that the total time taken is equal to $T_a + T_t$. Where $T_a = T_s + T_r$, the seek time and the rotational delay T_t is obtained using the expression in the previous slide. With that, we have a final value of 6.048ms

One point, notice that $T_a \gg T_t$ in this case.

What this means is that if the data of one file is spread across different sectors in different track, then you'll find that the effective HDD transfer rate will be much lower compared to the case where the data are all within the same track. This is because every time the RW head completes reading one sector, it has to move to the start of the next sector which is located in a different track, and that is going to take another T_a duration.

Not sure if you people still do disk defragmentation or not, but that is a process where you get the computer to organise the data so that data from the same file is allocate to consecutive sectors/track. Effect is that the HDD transfer rate should increase compared to before defragmentation.

Physical Layout

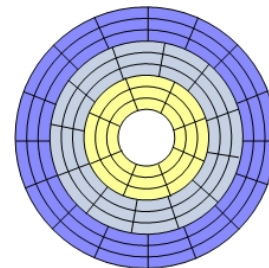


- Early HDD

- Physical Layout refers to the actual layout design on the HDD.
- Equal number of sectors per track and each sector has the same data size.
- Same-numbered Tracks from different surfaces formed a cylinder.
- The early hard disks were implemented using this topology to simplify the controller design.

- Modern HDD

- Having equal number of sectors per track means that sectors at the outer tracks are wider.
- Waste physical space as bit density of those sectors are not optimal.
- Zone bit recording technique addresses space wastage.
- Tracks are divided into zones, with differing number of sectors per track for different zones.



Physical layout refers to how data is organized and formatted on the HDD media.

In the past,

Every track has equal number of sectors and each sector contain the same amount of data.

This constant sector per track type of layout is similar to what you saw in earlier slides when we do the access time computation.

Early HDD design used this type of formatting for both logical and physical layout to simplify the controller design.

But the physical layout of the HDD today has adopted a slightly different format.

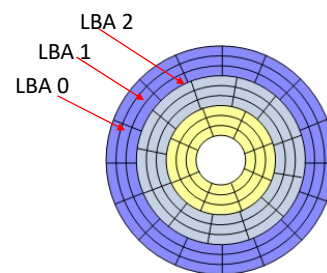
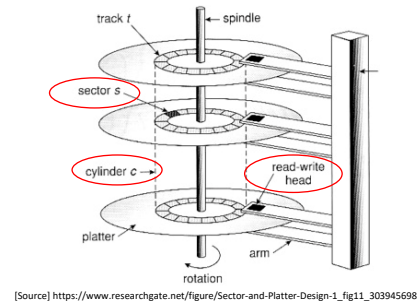
Primary reason is that the old method of formatting results in a lot of space wastage since the sector length on the outer track are much larger than those in the inner tracks, meaning that they could potentially store more data.

Hence HDD today uses the zone bit recording formatting. Where the sector length is kept approximately the same to fully utilize the recording media.

Outcome of this is that the number of sectors on the outer track is now more than those on the inner tracks.

Logical Layout

- How **software** sees & addresses HDD data?
- **Address translation**: needed to map the physical to logical locations. Two addressing scheme: **CHS** and **LBA**.
- **Cylinder-Head-Sector (CHS)**
 - Legacy scheme using the old HDD physical structure.
 - Made **obsolete** in recent standards due to addressing limitation & more **complex** formatting.
- **Logical Block Addressing (LBA)**
 - Simple **linear** addressing starting from LBA=0 as first block, LBA=1 as second block and so on.
 - 48-bit LBA standard allows addressing up to 128PByte. 1PByte=2⁵⁰ Bytes.



Logical layout and the corresponding addressing are what your program code uses when accessing the HDD.

There are two types of logical layout addressing used, CHS, which stands for Cylinder-Head-Sector, and LBA, short form for Logical Block Addressing.

CHS is the legacy scheme that uses the old constant sector per track layout and formatting convention i.e. Cylinder, Head and Sector.

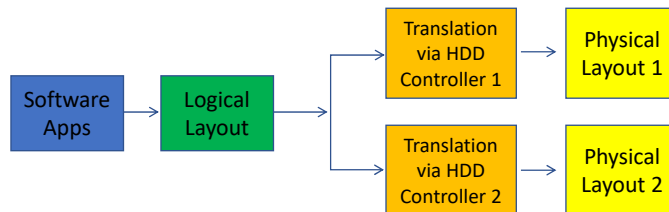
This scheme is obsolete these days due to limitation in the HDD capacity it can support.

CHS is replaced by the LBA, which uses a simple linear addressing scheme.

As shown in the diagram, the first block in the HDD is given the address LBA0, next block = LBA1 and so on until the last block of the HDD.

48 bits is allocated to specify the LBA number so this scheme have a lot of headroom, since it can address up till 2⁵⁰ bytes.

Logical vs Physical Layout



- Logical layout is needed in order to provide a common interface to all software developer
- If only physical layout is available, software has to be tuned for every HDD that uses a different physical layout
- Having a standard such as LBA allows the software developer to write application that can be used in any HDD.
- HDD vendor will provide the necessary firmware in their controller to perform the logical to physical layout address translation.

So why do we need two different layout format?

If only physical layout is available, a piece of software has to be tweaked every time it needs to interface to a new HDD.

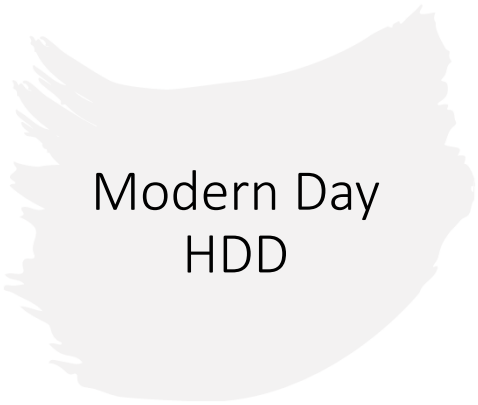
Because there is not standard way of designing a physical layout.

In fact, the physical layout of HDD from different vendors will be different, and is a way these vendors could differentiate their products and make money.

So in order to resolve this issue, we need a standard way for a software program to talk to the HDD, and that is where logical layout comes in.

With a standard such as LBA, software only needs to communicate according to this standard.

Whenever a HDD vendor develop a new product, they will need to ensure that proper translation of logical to physical layout can be done via their HDD controller.



Modern Day HDD

- Detail physical layout not given due to complex zone bit recording.
- Only the average transfer rate is given these days.
- Use of Cache and Buffers to speed up data transfer.
- However, concept and limitation of the magnetic HDD's physical design still valid
 - Actuator ARM is fixed and access has to be sequential, once data is missed, it has to wait for a disc revolution.
 - Seeking from track to track takes time as the R/W head needs to align to the starting sector.
 - More efficient to read in blocks rather than random access
 - Vulnerable to motion

If you buy any HDD these days, you will realise that not much information of the HDD is provided beyond the capacity of the HDD, RPM value and the Average Data Transfer Rate.

The detail physical layout information is not given because of the complexity of Zone Bit Recording, giving detail physical layout information will cause more confusion (than providing useful information).

The HDD these days uses Cache and Buffers to speed up the data transfer.

However, the concepts and limitation we learn here is still applicable

The mechanics of the HDD and its associated limitation remain, e.g. the actuator ARM is fixed and access has to be sequential.

Seeking from track to track consumes more time

So it is more efficient to read/write in blocks rather than random sectors

Which means HDD defragmentation is still applicable and helps to improve performance

And HDD, due to its mechanical nature, is vulnerable to motion.



Solid State Drive (SSD)

- Solid-state drives (SSD) are becoming popular
 - Memory array based on NAND-FLASH or NOR-FLASH.
 - Still more expensive than magnetic hard-disk.
- Flash memory has limited program-erase cycles (3,000 to 1,000,000).
- Various techniques used to extend the life of SSD
 - Wear levelling is a technique used to extend the life of the SSD disk by distributing data erase/write operations evenly over the entire disk.
 - Use External RAM as buffer to minimize the number of writes to Flash in SSD.
 - Error Correction Code. Ability to recover from one or more bits of error in media.
- MTBF (Mean Time To Failure) of recent SSD is comparable to that of HDD.

Next we come to the solid state drives.

These drives are getting more and more popular as they are more robust to handling, since they don't have any moving parts compared to magnetic HDD. And also, primarily because the cost of NAND flash memory, which is the main memory used in SSD, has dropped drastically over the last few years.

It is, however still more expensive than magnetic HDD so many of the computers, especially the desktops and servers, are still using magnetic HDD. HDD is also still the main storage element found in Data Centers.

One of the main issues with SSD is the limited program-erase cycles of flash memory that it uses.

What it means is that there is only limited number of times user can perform erase operation on a particular memory block location.

But there are many techniques to extend the life of SSD

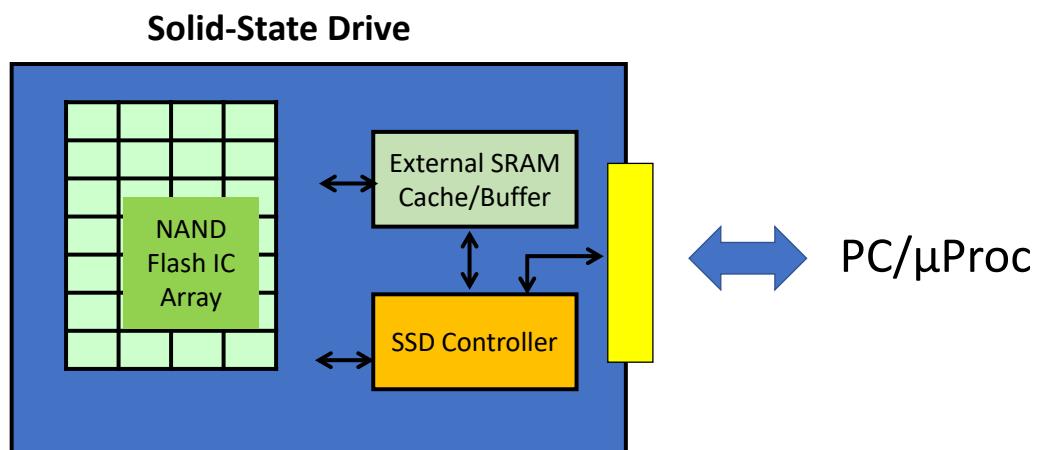
Wear levelling which spreads out the write/erase operation evenly over the entire

disk

Using external RAM as buffer to reduce the number of memory access to Flash memory.

With these enhancement and mitigation, the mean time to failure benchmark of SSD these days are comparable to magnetic HDD.

SSD System Block Diagram



This slides shows the system block diagram of a SSD.

The main storage elements are NAND flash.

There is a SSD controller which a basically a microprocessor that manages the operation of the SSD, and this includes:

Addressing translation between Logical to Physical Layout

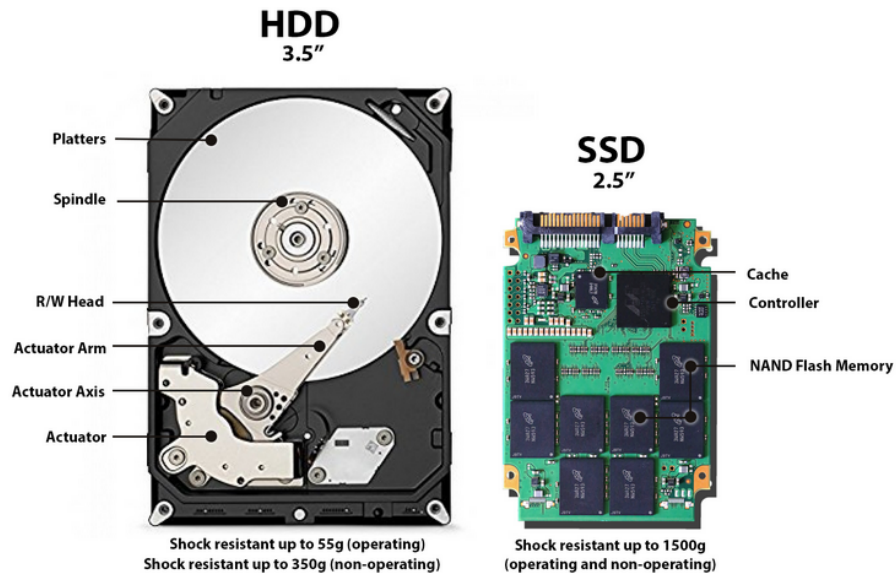
Caching and buffering of data in and out of the SSD

Wear levelling algorithm

There are external SRAM and Cache to improve the transfer rate for both read and write operation

So in a way, SSD itself is a computer system.

HDD vs SSD



Source: <https://www.backblaze.com/blog/ssd-vs-hdd-future-of-storage/>

- This is a photo comparison between a HDD and a SSD.
- form factor: a hardware design aspect that defines and prescribes the size, shape, and other physical specifications of components
- SSD form factor can be made smaller actually, such as the mSATA and NVMe form factor which is much smaller than the legacy 2.5inch HDD form factor.
- The earlier SSD adopted the 2.5 inch form factor so that they can be a drop-in replacement for the 2.5inch HDD found on notebook computers.

HDD vs SSD

- HDD

- Pros

- Lower Cost Per Bit,
 - Almost infinite Erasure cycles

- Cons

- Consist of Moving Mechanical Parts so more prone to crashing if HDD is dropped or shaken.
 - Heavier and larger physical profile.
 - Slower transfer rate

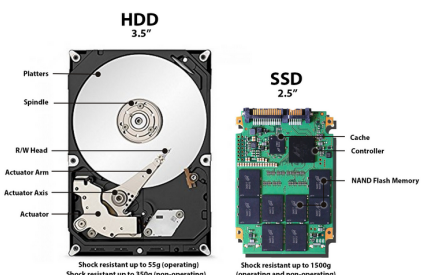
- SSD

- Pros

- No moving parts. More robust to movement.
 - Lighter and occupy less space
 - Higher Transfer rate

- Cons

- More Costly compared to HDD
 - Finite number of Erasure cycles



Source: <https://www.backblaze.com/blog/ssd-vs-hdd-future-of-storage/>

Last slide of the HDD/SSD video.

Advantage of the HDD

It has lower cost per bit and
Almost infinite erasure cycles.

Disadvantage of HDD

There are many moving parts in HDD, and it is therefore prone to crashing if there are huge movements

Its form factor is larger and more bulky
a slower transfer rate.

Advantage of SSD

Opposite of HDD, there are no moving part so it is very robust against movement.
It is very light; can be made really small and
Has a higher transfer rate

Disadvantages of SSD

Cost. Its more expensive cost per bit wise when compared with HDD

There is a limit to the number of erasure cycles user can apply on the SSD.

We have come to the end of this video, and next video will be on Data Center related information.

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